

Report Biomimicry Challenge

ZHAW Institute for Sustainable Development | Blockweek Bionics 2026

Challenge 1 — Urban Heat Islands

Cooling the City



Concept V2 on the reference block: timber ribs and collector panels on the façade (render: project team)

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Contents

1	Introduction	2
1.1	Background & Relevance	2
1.2	Design Challenge & Project Contribution	2
2	Challenge & Scoping	3
2.1	Problem Definition	3
2.2	Context and Constraints	3
2.3	System View and Project Boundary	3
2.4	Design Question	5
2.5	Functions and Functional Diagram	5
2.6	Biologized Questions	6
2.7	Performance Criteria and KPIs	6
3	Bio-Inspired Design Process	8
3.1	Discovering & Abstracting	8
3.2	Creating: Engineering Concept Development	13
3.3	Concept Evaluation & Iteration	19
4	Sustainability & Life's Principles	24
4.1	Life's Principles applied to Concept V2	24
4.2	Environmental benefits, limitations and trade-offs	24
4.3	Materials, durability, repair and circularity	25
4.4	Did sustainability considerations change the design?	26
4.5	Limits of this assessment	26
5	Discussion & Conclusion	27
5.1	Contribution of the biomimicry process	27
5.2	Strengths and weaknesses of the final concept	27
5.3	Assessment of the challenge and the KPIs	27
5.4	Innovative character and limits of the evidence	27
6	References	28
	Appendix A – AI Use Statement	31
	Appendix B – Optional Supporting Material	32

1 Introduction

1.1 Background & Relevance

The Urban Heat Island (UHI) effect represents a growing environmental and public health challenge, characterized by significantly elevated temperatures in urban environments relative to surrounding rural areas [1]. This effect is worsened by climate change and a high concentration of heat absorbing infrastructure, like cement and asphalt. Conventional mitigation strategies rely heavily on air conditioning, which creates a counterproductive cycle (Section 2.1).

1.2 Design Challenge & Project Contribution

The design challenge lies in developing an efficient, scalable cooling system capable of mitigating the urban heat island effect without relying on high energy consumption. The goal is to analyze the different heat dissipation methods found in nature, specifically in animals, to find a solution which conforms with the given challenges.

2 Challenge & Scoping

2.1 Problem Definition

Conventional cooling methods such as air conditioning worsen the Urban Heat Island effect by displacing the warm air from inside the building to the surrounding area. This local heat exchange causes the city to warm up more, increasing the need for air conditioning, which ultimately results in a self-reinforcing feedback loop.

2.2 Context and Constraints

All context values, benchmarks and engineering checks in this report refer to the City of Zurich. Zürich was chosen for four reasons:

1. The city has adopted a heat-mitigation strategy, the Fachplanung Hitzeminderung [2], with an implementation agenda, a guidance leaflet for building owners and thematic maps. The project therefore has a documented heat map and an albedo classification to refer to.
2. The city publishes an open climate analysis including geodata, which let us select a documented reference site.
3. MeteoSwiss operates long term measurement stations in and around the city.
4. Its building style and infrastructure are representative of the Swiss Mittelland, so results transfer to comparable Swiss cities.

2.2.1 Reference Building

For comparison and reference purposes a building inside of Zurich was chosen. The chosen building is the Modellierungsgebiet 03 – Geschlossene Randbebauung (closed perimeter block). It consists of two outward-closed blocks of five stories including a roof structure and an enclosed private courtyard. It is located in a dense district close to the city center. The city's assessment is that the reduced air exchange and lack of cold air supply makes this building overheat comparatively quickly [2, pp. 148–151].

2.2.2 Constraints

Table 2.1 separates hard constraints where a violation disqualifies a concept, from soft constraints which weight the evaluation but do not necessarily disqualify a concept.

2.3 System View and Project Boundary

2.3.1 Super-system

The chosen super system is the city and its existing infrastructure.

2.3.2 System of interest

The core interest of this project focuses on heat-regulation of the outer layer of a building including the façade and the roof.

Table 2.1: Context conditions and constraints.

Type	Class	Constraint
Energy	hard	Limit auxiliary electrical energy for the cooling function.
Water	hard	No unnecessary water waste.
Townscape	hard	The result must be acceptable in the Zurich townscape. Buildings may be listed; heritage protection, building law and neighbor rights can restrict visible changes [2, p. 128].
Glare	hard	No specular glare towards streets, neighboring flats or traffic, which can impede day to day life.
Climate	hard	Must survive the Swiss annual cycle: frost to about -10°C , freeze-thaw cycles, driving rain, UV-Light, and periods of several dry weeks in summer (team design assumptions).
Winter	hard	Must not increase heating demand. A high solar reflectance also reduces useful winter solar gains; this trade-off has to be quantified.
Existing building	soft	Preferable, but not necessary. Concepts can also be developed for only new buildings.
Installation	soft	Must be installable by ordinary trades on an occupied building, ideally without full scaffolding.
Maintenance	soft	Must remain functional with little maintenance.
Solar energy	soft	Should not unnecessarily compromise roof- or façade-integrated photo-voltaics.
Structural	soft	Limited additional dead load and build-up thickness on an existing envelope.
Embodied emissions	soft	The greenhouse gas embodied in the added infrastructure must be repaid by the avoided cooling demand within the service life, measured in kg CO ₂ -eq per m ² .
Economic	soft	Cost per square meter should be in the range of a normal refurbishment.

2.3.3 Parallel systems

- Stormwater and sewer network
- Street and pavement network
- Power grid and building energy system
- Residents
- Cars, Bicycles, Public transport and other forms of transportation

2.3.4 Subsystems

- Outer surface
- Storage and insulation layer
- Water paths
- Attachment to the substrate

Table 2.2 states which aspects are scoped in or out.

Table 2.2: Project boundary: what is deliberately in and out of the project.

IN the project	OUT of the project
Prevent heat buildup and remove heat from the building façade.	Redesign of the building itself. Reuse of the stored energy. Internal cooling.

2.4 Design Question

2.4.1 Candidate questions considered

Table 2.3 shows the rejected candidates.

Table 2.3: Design questions considered, and why they were rejected.

#	Candidate	Assessment
1	How do we prevent heat build-up in urban areas?	Too broad. No place, no time, no intervention point. The equivalent of “How might we end world hunger?”
2	How could we use the energy or heat build-up positively while lowering temperature?	Right idea, but unanchored: no location, and “positively” does not say for what.
3	How might we get the heat that Zurich’s existing buildings store during the day back out of the city before the next morning, without using energy?	Well-formed and highly testable, but it prescribes when and how the heat has to leave and leaves little room for variation. Its intervention point, the existing buildings, is taken over into the selected question.
4	How can we prevent heat waves causing heat islands?	Factually wrong. Heat waves do not cause heat islands. The urban heat island is a permanent consequence of urban infrastructure and exists on ordinary summer days. A heat wave makes it dangerous, but it does not create it.

2.4.2 Selected design question

How might we reduce the urban heat island effect in cities like Zurich by changing the façade and roof of existing buildings?

The added clause fixes the intervention point at the system of interest of Section 2.3. It allows some freedom in how to reduce the urban heat island effect, and it takes over the mechanism of candidate 3 without prescribing when and how the heat has to leave.

2.5 Functions and Functional Diagram

2.5.1 Functions

Table 2.4 displays the chosen functions and their respective taxonomy link.

Table 2.4: Functions list.

ID	Function (verb)	Taxonomy link – Explanation
F1	Reflect short-wave radiation	Prevent solar radiation – Avoid absorbing solar energy (Visible & near infrared light).
F2	Shade a surface	Prevent excess temperature – Reduce direct light exposure on structural surfaces.
F3	Dissipate / Transfer heat	Modify thermal conditions – Transfer thermal energy via transmission or thermal radiation.
F4	Evaporate liquid	Regulate physical processes – Heat extraction using moisture or water
F5	Resist degradation	Prevent structural breakdown – Maintain long-term durability against UV, rain, and frost.

2.5.2 Functional Diagram

Figure 2.1 draws the functions of Table 2.4 as a flow picture. Incoming shortwave radiation is either reflected (F1) or the surface is shaded (F2); what is absorbed raises the surface temperature and leaves again by F3, to the sky and to the street-canyon air, or by F4, where liquid water enters the layer and leaves as vapour. Whatever neither path removes flows into the building. The surface temperature is the signal that sets how strongly F3 and F4 act.

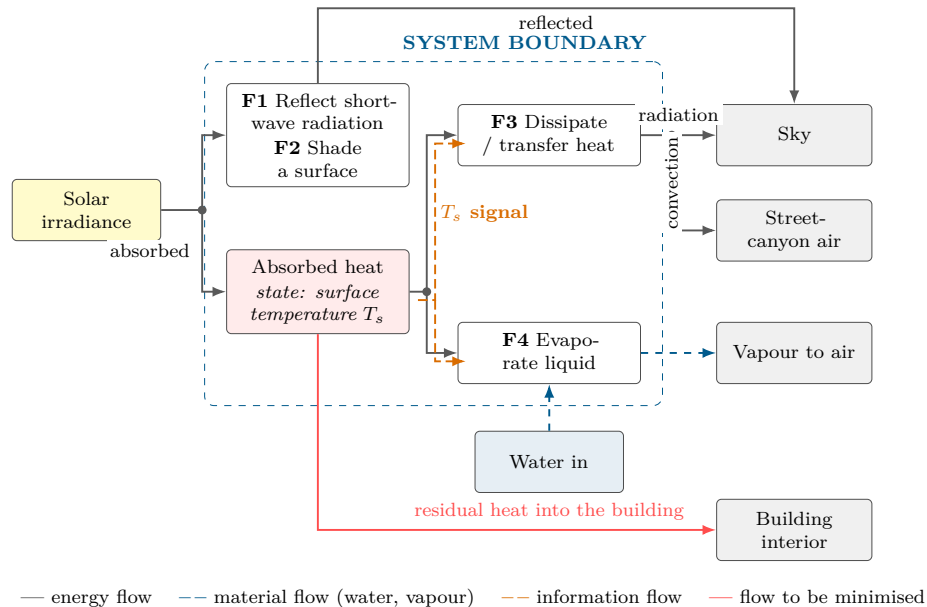


Figure 2.1: Solution-neutral functional diagram; the system boundary is the outer layer of the building. Solid arrows are energy flows, blue dashed arrows the material flow of F4 (water in, vapour out), the orange dashed arrows the information flow: the surface temperature drives the strength of F3 and F4. F5 (resist degradation) acts on everything inside the boundary and is not drawn.

2.6 Biologized Questions

1. How does nature reflect incoming shortwave radiation?
2. How does nature shade a surface even in the desert?
3. How does nature dissipate excess heat?
4. How does nature transfer excess heat?
5. How does nature dissipate excess heat using moisture?
6. How does nature keep a functional surface clean without cleaning it?

2.7 Performance Criteria and KPIs

2.7.1 KPIs

The Albedo index defines a value between 0 – 1 where 0 is total absorption of all incoming radiation and 1 is the total reflection of all incoming radiation [3]. Shading provides an alternative way to keep a surface cool [4]. Water evaporation is an effective solution to remove heat using moisture as means of heat transfer; the aim is to reduce the needed amount of water evaporation to conserve water.

Table 2.5 defines one KPI per function, with the unit in which we compare concepts, the baseline of the untreated reference block, the target, and the measurement we would use to check it. The baseline class and the target of KPI1 come from the albedo classification of the

Fachplanung, which rates 0.30 and below as low and 0.80 and above as high [2, p. 128]. The target of KPI5 follows from the Water constraint in Table 2.1. The targets of KPI2, KPI4 and KPI6 are team working targets: we set them to have a number to test against, and they are not city values. KPI3 and KPI4 both belong to F3 but measure different things: KPI3 counts the heat that actually leaves the façade through a removal path, which is the quantity the engineering checks in Section 3.2.7 and Section 3.2.8 estimate; KPI4 is the effect on the surface that a pedestrian is exposed to.

Table 2.5: KPIs. Targets marked “team” were set by the team; the KPI1 target and baseline class come from the Fachplanung [2, p. 128].

#	Fn	KPI	Unit	Baseline	Target	Measurement
KPI1	F1	Solar reflectance of the exposed surface	– (0–1)	Existing render and roof of the reference block, class “low” (≤ 0.30)	≥ 0.80 , the city’s “high” class	Spectrophotometer or pyranometer pair, new and after weathering
KPI2	F2	Share of the south and west façade area shaded between 12:00 and 18:00 on the design day (21 June, clear sky)	%	0 % (bare façade)	≥ 70 % (team)	Shadow simulation on the 3D model, then site photos
KPI3	F3	Heat removed from the façade and released outside the street canyon	kW total; W/m ² of connected façade	0 (no removal path today)	$\geq$ the collected façade load, taken as 100 W/m ² in Section 3.2.7	Calorimetric: flow $\times c_p \times \Delta T$ on the loop; for a coating, surface heat-flux plates
KPI4	F3	Reduction of the façade surface temperature at 18:00 on the design day, against the untreated reference façade	K	0 K	≥ 5 K (team)	Infrared thermometer on treated and untreated bays of the same façade
KPI5	F4	Water consumed in operation	L per m ² façade per day	0 (the reference façade uses none)	0 (Water constraint, Table 2.1)	Water meter on the make-up line
KPI6	F5	Performance retained after 20 annual cycles, for KPI1 and KPI3	% of initial	–	≥ 90 % (team)	Accelerated weathering (UV, freeze–thaw, driving rain), then repeat KPI1 and KPI3

2.7.2 Life Principles

The life principles which should reflect the end solution include:

- Be resource efficient: minimize the use of external resources
- Adapt to changing conditions: adapt to day/night time as well as season cycles
- Use life-friendly chemistry: refrain the use of toxic chemicals or materials

3 Bio-Inspired Design Process

3.1 Discovering & Abstracting

3.1.1 Research path

To find suitable solutions both the AskNature website as well as the AskNature chat were used. The following keywords were used to find solutions produced in the nature: cooling, thermal transfer, shading, heat reduction, reflect, evaporation. Additionally the biologized questions were used to search for solutions using search engines such as Google.

The models that were selected had to fulfill at least one function mentioned in Section 2.5. Additionally they had to fit into at least one of the defined life principles. Table 3.1 lists every candidate we looked at, the nine models we kept and the four we dropped, with the reason for each decision.

Table 3.1: Candidate biological models, the functions and Life's Principles they serve, the quality of the evidence we found, and the selection decision.

Organism / model	Fn.	Life's Principle	Evidence	Decision and reason
<i>Selected</i>				
African elephant, ears	F3	Resource efficient; adapt to changing conditions	AskNature page only [5]; same strategy in jackrabbits [6]	Kept. Separates the place where heat is collected from the place where it is released.
African elephant, sparse skin hairs	F3	Resource efficient	Primary paper [7]	Kept. Heat transfer gain measured at the low wind speeds that occur in a street canyon.
Cactus ribs	F2, F3	Resource efficient	Primary paper [8]; AskNature [9]	Kept. Geometry only, no material, no water.
Saharan silver ant, hairs	F1, F3	Adapt to changing conditions; life-friendly chemistry	Primary paper [10]; AskNature [11]	Kept. The only model that reflects sunlight and raises infrared emission in one structure.
Desert snail <i>Sphincterochila boissieri</i> , shell	F1, F3	Resource efficient; adapt to changing conditions	Primary paper [12]; AskNature [13]	Kept. A passive mineral surface with measured reflectance.
<i>Morpho</i> butterfly, wing scales	F1	Resource efficient	Primary paper [14]; AskNature [15]	Kept with a limitation. Shows how structure can replace pigment; it does not show near-infrared rejection.
Dromedary camel, fur and sweat	F2, F4	Resource efficient; adapt; life-friendly chemistry	Primary paper [16]; AskNature [17]	Kept. Shows how insulation reduces the water needed for evaporative cooling.
Pine cone scales	F2	Adapt to changing conditions; life-friendly chemistry	Primary paper [18]	Kept. A passive actuator driven by the weather.
Western honeybee, fanning	F3	Resource efficient; adapt to changing conditions	Primary papers [19], [20]	Kept. Shows how many simple elements give a graded response to load.
<i>De-selected</i>				

Organism / model	Fn.	Life's Principle	Evidence	Decision and reason
Termite mound	F3	—	Mechanism contested in the literature	Dropped. The chimney account of mound ventilation is disputed, and the building usually cited for it uses fans.
Namib desert beetle	—	—	Well documented fog harvesting	Dropped. Water capture lies outside the project boundary (Table 2.2), and a Zurich heat wave brings neither fog nor rain.
Thomson's gazelle, carotid rete	F3	—	Internal counter-current exchange	Dropped. Internal physiology with no surface to transfer to a building envelope.
Green façades and roofs	F2, F4	—	Established building measure	Dropped. Outside the boundary and in competition with the water constraint: planting needs most water when least is available.

3.1.2 Natural Models

African Elephant: Ears **Biological mechanism.** As seen in Figure 3.1 African elephants have large and highly vascularized ears which they use as heat exchange surfaces. Warm blood is transported from the body core into the ear, where heat is transferred to the surrounding air. By flapping their ears additional airflow is created which further increases the heat transfer [5]. This same approach is also used by the jackrabbit [6].

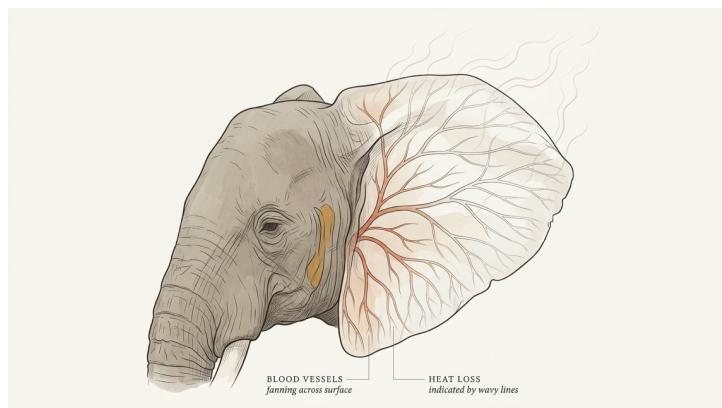


Figure 3.1: African Elephant ears. Source: [TODO: image source].

Functional abstraction. Transfer thermal energy from a heat generating region to a large, exchange surface and increase airflow across this surface.

Design principle for urban cooling. Use dedicated heat rejection surfaces in locations with favorable airflow to transfer and dissipate heat from urban structures.

Life's Principle: Be resource efficient / Adapt to changing conditions

Elephant Skin: Sparse Hairs Biological mechanism. African elephants have additionally to their ears a second cooling mechanism. They have very sparsely spread hair all over their body. Due to the low density of the body hair it does not act as an insulation layer but rather as additional surface area to transfer heat. As seen in Figure 3.2 it also serves to break the stationary air surrounding the skin. This allows for a better heat transfer; Myhrvold et al. measured up to 23 % more heat loss at low wind speeds [7].

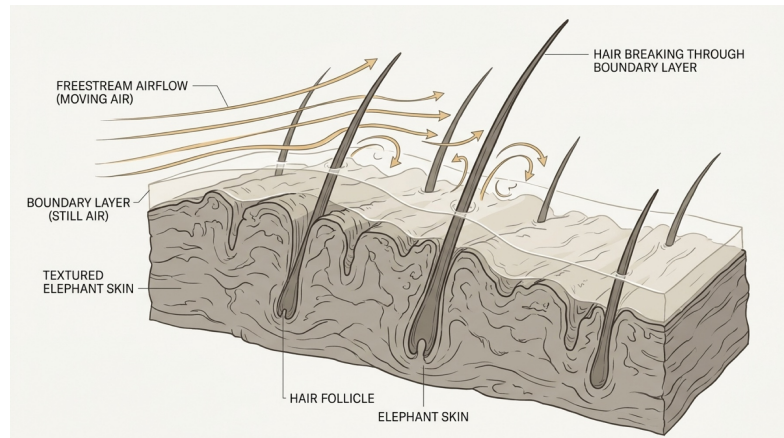


Figure 3.2: African elephant hair. Source: [TODO: image source].

Functional abstraction. Increase heat-transfer area by distributing sparse bulges across a warm surface.

Design principle for urban cooling. Use small surface structures on façades or other heated urban surfaces to enhance passive heat transfer without substantially increasing material use.

Life's Principle: Be resource efficient

Cactus Ribs Biological mechanism. Many cacti have ribbed surfaces consisting of repeated peaks and troughs. The peaks shade parts of the troughs and therefore reduce direct solar exposure. Air within the shaded troughs remains cooler. The ribbed geometry also increases surface area and disrupts airflow around the plant, similar to the elephant hair [8], [9].

Functional abstraction. Use folded surface geometry to combine self-shading, increased heat-transfer area and controlled interaction with airflow.

Design principle for urban cooling. Integrate ribbed or folded geometries into urban surfaces to reduce direct solar exposure while simultaneously promoting passive heat dissipation.

Life's Principle: Be resource efficient

Saharan Silver Ant: Reflective Hairs Biological mechanism. The Saharan silver ant is covered with densely packed hairs that have a triangular cross-section and grooved surfaces. These hairs reduce heat gain in two ways: they strongly reflect visible and near-infrared solar radiation through total internal reflection and they increase emissivity in the mid-infrared range, allowing the ant to radiate excess body heat more efficiently. Shi et al. measured that ants with hairs stay about 5–10 °C cooler than ants whose hairs were removed [10], [11].

Functional abstraction. Use micro structured surfaces to reflect incoming solar radiation while simultaneously enhancing thermal radiation away from the surface.

Design principle for urban cooling. Develop façade or roof surfaces with micro structured geometries that minimize absorption in the visible and near-infrared solar spectrum while maximizing heat emission in the mid-infrared. This could reduce solar heat gain and improve passive radiative cooling without requiring active energy input.

Life's Principle: Adapt to changing conditions / Use life-friendly chemistry

Desert Snail Shell Biological mechanism. The desert snail *Sphincterochila boissieri* limits heat gain through a highly reflective shell and internal insulating air spaces as seen in Figure 3.3. Schmidt-Nielsen et al. measured approximately 90 % reflectance in the visible spectrum and about 95 % in the near-infrared range [12]. The snail also withdraws into the upper parts of its shell, while air-filled spaces and limited contact with the hot substrate reduce conductive heat transfer [13].

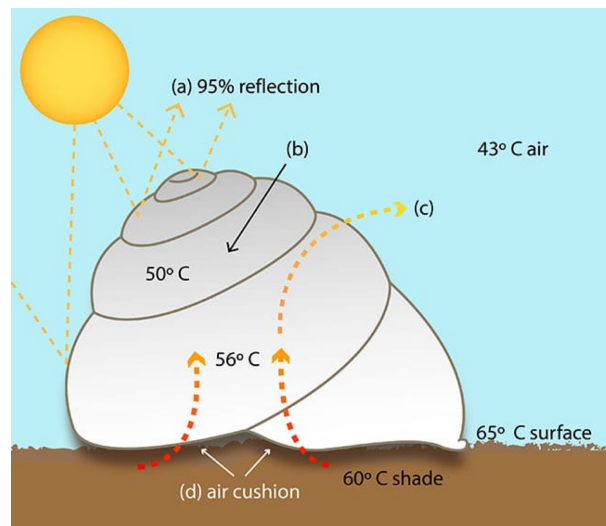


Figure 3.3: Desert Snail [13].

Functional abstraction. Reduce incoming radiant energy and interrupt conductive heat transfer using reflective surfaces and stationary air layers.

Design principle for urban cooling. Combine highly reflective external surfaces with insulating or ventilated air cavities to limit solar heat storage in roofs and façades.

Life's Principle: Be resource efficient / Adapt to changing conditions

Morpho Butterfly Wing Biological mechanism. Morpho butterflies generate their characteristic color using transparent chitin-and-air structures at micro- and nanoscale. As seen in Figure 3.4, ridges and cross-ribs interact with incoming light through diffraction and interference, causing selected wavelengths to be strongly reflected. The color therefore results primarily from physical structure rather than pigmentation [14], [15].

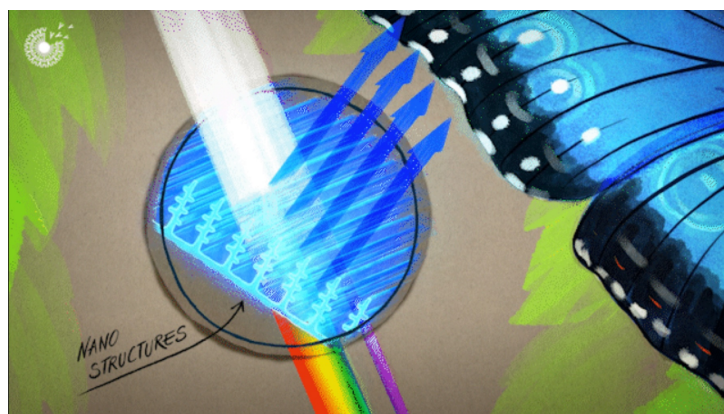


Figure 3.4: Reflection with nano structures [14].

Functional abstraction. Control the spectral interaction between a surface and incoming radiation through micro- and nanoscale geometry.

Design principle for urban cooling. Engineer structured surfaces that selectively manipulate incident solar radiation while maintaining desired optical properties. The Morpho demonstrates structural color rather than near-infrared heat rejection, and the cooling application requires separate verification.

Life's Principle: Be resource efficient

Camel Fur and Sweat Biological mechanism. Dromedary camels combine insulation with evaporative cooling. Figure 3.5 displays how exactly this mechanism works. The sweat glands release water which removes the body heat while evaporation and the thick fur reduces the heat that reaches the skin, so that less sweat is needed [16]. Furthermore the camel reflects light with their pale color [17].

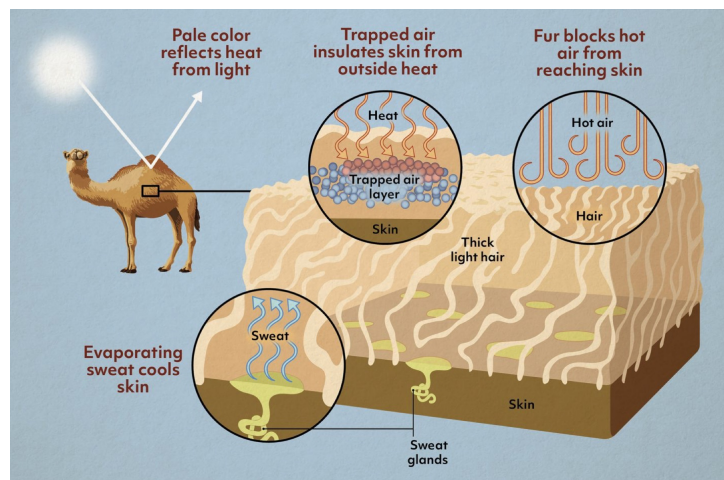


Figure 3.5: Insulation with evaporative cooling [17].

Functional abstraction. Block heat using insulation in combination with evaporation as a means of heat dissipation.

Design principle for urban cooling. Combine shading or thermal insulation with controlled evaporative cooling so that less water is required to achieve the desired cooling effect.

Life's Principle: Be resource efficient / Adapt to changing conditions / Use life-friendly chemistry

Pine Cone Scales Biological mechanism. Pine cone scales respond to humidity despite containing no muscles or active control system. Their layered structure contains materials with different swelling behavior. When moisture is absorbed, one layer expands more than the other, causing the scale to bend and close. As the scale dries, it returns towards its open state. The movement is reversible [18].

Functional abstraction. Convert an environmental change directly into mechanical movement through differential material expansion.

Design principle for urban cooling. Use environmentally responsive material layers to create self-actuating shading or ventilation elements that operate without motors or electronic control.

Life's Principle: Adapt to changing conditions / Use life-friendly chemistry

Honeybee Fanning Thresholds Biological mechanism. Western honeybee colonies regulate nest temperature through the combined behavior of many workers. Individual bees have different temperature-response thresholds [20]. As nest temperature increases, progressively more bees begin ventilating the hive. This distributed response prevents excessive system-wide reactions to small temperature fluctuations and stabilizes the nest with no central control [19].

Functional abstraction. Distribute different activation thresholds across multiple independent elements to create a gradual system-level response.

Design principle for urban cooling. Use multiple passive cooling elements with different trigger conditions so that cooling capacity increases progressively with thermal load instead of operating through a single on/off threshold.

Life's Principle: Be resource efficient / Adapt to changing conditions

3.2 Creating: Engineering Concept Development

3.2.1 Structured Ideation: Crazy 8s

Table 3.2 translates the biological mechanisms in Section 3.1 into eight technical building blocks.

Table 3.2: Crazy 8s building blocks, before concept combinations.

Idea	Fn.	Biological starting point	Technical building block
1 Fixed shade	F2	Cactus ribs change exposure.	Projecting ribs shade the façade.
2 Conductive fins	F3	Elephant hairs act as small fins.	Separated projections conduct heat to air.
3 Surface texture	F3	Sparse hairs suggest distributed structures.	Small roughness elements alter near-wall airflow.
4 Solar reflector	F1	Silver-ant hairs reflect sunlight.	A reflective coating reduces solar absorption on a façade or roof.
5 Infrared emitter	F3	Silver-ant hairs enhance thermal emission.	A sky-facing surface releases infrared heat.
6 Fluid transport	F3	Blood carries heat to jackrabbit ears.	A fluid carries heat to a separate exchange surface.
7 Variable exchange area	F3	Ear blood flow regulates heat release.	Connect more exchange area when cooling demand rises.
8 Responsive shade	F2	Unequal swelling bends pine-cone scales.	A humidity-responsive layer changes shade position.

3.2.2 Combining Ideas with a Morphological Matrix

Table 3.3 groups alternatives by function. Bracketed numbers refer to Table 3.2; Option 3 draws on further models from Section 3.1.

Table 3.3: Alternatives for heat shielding, reflection, heat transport and heat release.

Design choice	Option 1	Option 2	Option 3
S – Heat shielding	Fixed ribs (1)	Responsive louvers (8)	Insulating outer layer (camel fur)
O – Solar reflection	Reflective roof (4)	Reflective façade (4)	Multilayer reflector (Morpho wings)
T – Heat transport	Fluid loop (6)	Conduction through texture (2/3)	Fan-driven air cavity (bee fanning)
E – Heat release	Roof emitter (5)	Wall exchange with local air and sky	Evaporating wet layer (camel sweat)

A = S1–O1–T1–E1 combines fixed fins, a reflective roof and a water loop. **B = T2–E2** uses a textured coating for local heat exchange. **C = S2–O2–E2** combines responsive louvers with a reflective façade. Variable connected area (7) remains an unselected regulation option for **A**.

Camel fur limits heat gain and sweating removes heat by evaporation, suggesting an insulating outer layer and a wet cooling surface [16]. Morpho wing structures suggest a multilayer reflector, although solar cooling requires separate verification [14]. Bee fanning suggests temperature-switched fans moving air through a cavity [19]. These options were not developed further because of water or power demand, uncertain optical performance, or insulation restricting outward heat transfer.

C was set aside because humidity does not directly indicate cooling demand. Reflective façades also increased pedestrian heat stress in simulations of a Stuttgart street canyon [21]. **A** therefore places its reflective surface on the roof. **A** and **B** were retained for the engineering comparison below.

3.2.3 Concept A V1 — Water Façade

A V1 was conceived for a new building, with water pipes in an absorber layer in front of the wall and timber fins providing shade (Figure 3.6). The water loop carries residual heat to a separate roof surface for release. V1 has no access panels in front of the pipes and the fins reach the ground; both points are changed in V2 (Section 3.3.3).

Cactus ribs inspire shading [8]; blood flow to jackrabbit ears inspires heat transport to a separate exchange surface [6]. Warmer water can rise and cooler water return if buoyancy overcomes pipe resistance [22].

3.2.4 Concept B V1 — Roughness Spray Retrofit

B V1 is a retrofit coating for an existing wall. Elephant hairs suggest small heat-transfer structures, but work as conductive fins [7]. The spray instead aims to disturb warm near-wall air (Figure 3.7). Its benefit depends on texture size, spacing, conductivity and local wind.

The build-up we propose is, from the wall outward, the existing rendered wall, a mineral primer for adhesion, and a textured top coat 2–4 mm thick that carries sparse protruding elements. The element geometry follows the density argument of Myhrvold et al. [7]: dense structures trap air and insulate, sparse ones act as fins. We therefore assume elements 5–15 mm high at a centre spacing of 20–40 mm. Candidate materials are a silicate or lime-cement binder with quartz or



Figure 3.6: Concept A V1. Timber ribs shade the façade; water carries residual heat to the roof and returns after cooling. Drawing: project team.

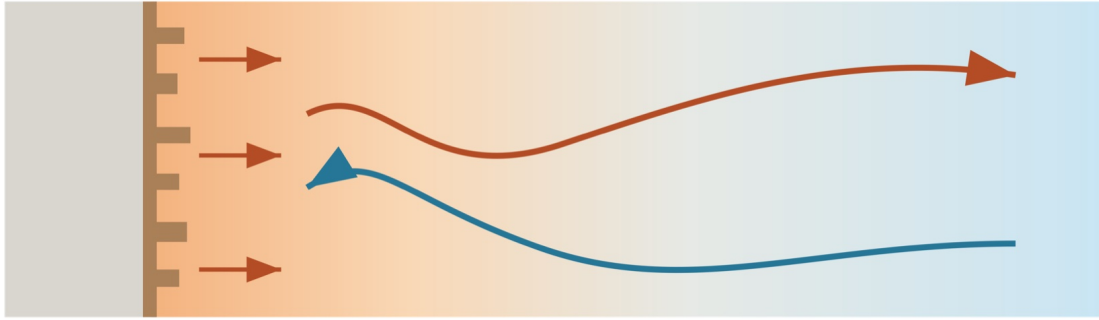


Figure 3.7: Proposed roughness mechanism. Orange: warm near-wall air (thermal boundary layer); blue: cooler surrounding air, with a gradual temperature transition. Straight arrows: heat from the wall. Curved arrows: proposed air exchange; its cooling benefit requires testing. Drawing: project team.

expanded-glass granules as the protruding elements. The conductivity of the granule matters less than the disturbance of the near-wall air, because the wall itself supplies the heat. The chemistry of binder and granules is open and belongs to the material checks of Section 3.3.6.

To give **B** a number of the same kind as **A** we made an order-of-magnitude estimate. On a vertical wall in still air the natural-convection coefficient is about $h \approx 3 - 5 \text{ W}/(\text{m}^2 \text{ K})$; façade coefficients in this range appear at low wind in the CFD study of Blocken et al. [23], and the order follows from the free-convection correlations in White [24]. Myhrvold et al. [7] measured up to 23 % more heat loss from sparse hairs at low wind. If the texture reaches the same enhancement, the added coefficient is $\Delta h \approx 1 \text{ W}/(\text{m}^2 \text{ K})$. The extra heat removed is

$$\Delta \dot{Q} = \Delta h A \Delta T \approx 1 \text{ W}/(\text{m}^2 \text{ K}) \times 1220 \text{ m}^2 \times 10 \text{ K} \approx 12 \text{ kW}, \quad (3.1)$$

with A the connected façade area of Table 3.5 and ΔT an assumed wall-to-air difference of 10 K on a hot afternoon. The estimate assumes still air; wind would raise both the base coefficient and the absolute gain. It also assumes the sign of the effect: Ahmed and Tanda [25] found that transverse ribs on a vertical surface can reduce laminar natural convection, so whether a given

texture helps or hurts depends on its geometry and must be measured. If the mechanism works, **B** could remove roughly a quarter of the 44 kW that **A**'s roof can release (Section 3.3.5), with a few millimetres of coating instead of pipes, panels and a roof plate. Section 3.3.1 uses this figure.

3.2.5 Engineering Canvas

Table 3.4: Engineering Canvas for the two V1 concepts.

Canvas item	A V1 — water façade	B V1 — roughness spray
Function	Prevent heat gain; transport and release residual heat.	Increase local heat exchange with a thin retrofit.
Biological principle	Cactus shading; blood-borne heat transport; silver-ant optical properties.	Sparse elephant hairs provide distributed heat-transfer structures.
Technical transfer	Shade, conduct heat into water, circulate by buoyancy, release it at a reflective roof emitter.	Conduct heat into a texture and test its effect on airflow.
Components / materials	Timber ribs, collector, pipes, roof plate/coating, insulated return, expansion vessel and vents.	Existing wall, coating binder and textured particles (parameters in Section 3.2.4); chemistry remains open.
Inputs / outputs	Sunlight and temperature difference → reflected light and roof heat release. Water recirculates.	Sunlight and airflow → heat exchange with local air and sky.
Requirements / KPIs	Section 2.7 KPIs; little auxiliary energy; winter operation, glare and townscape constraints.	Same KPIs; thin application, adhesion and safe weathering.
Critical assumptions	Enough heat reaches the water; roof capacity and buoyancy are sufficient.	Texture improves heat transfer without trapping still air.
Uncertainties	Pipe resistance, roof output, leakage, frost and panel-to-pipe contact.	Cooling benefit, soiling, adhesion and weathering.
Next check	Estimate circulation and roof capacity.	Compare smooth and textured samples under the same wind and temperatures.

3.2.6 Critical Engineering Questions

Q1: Can buoyancy circulate enough water? Q2: Can the roof release the collected heat? Q3: Does the spray improve heat transfer under weak airflow?

3.2.7 Engineering Check 1 — Circulation and Pipe Size

The first estimate applies **A V1** to a hypothetical new building with the dimensions of the Modellierungsgebiet 03 reference block. Table 3.5 lists the model inputs and engineering assumptions; the building dimensions were measured on our model of the reference block.

Table 3.5: Assumptions for the circulation estimate.

Input	Value and basis
Building and façade	64×58 m outer plan; 42×36 m courtyard; eaves at 15.4 m and roof rise 4 m. Two faces, assumed 65 % opaque: $\text{Area} = (64 + 58) \times 15.4 \times 0.65 \approx 1220 \text{ m}^2$.
Collected heat	Assume $q'' = 100 \text{ W/m}^2$ enters the water after shading; test 50–150 W/m^2 . Collection depends on panel-to-pipe contact.
Height H	Assume 10 m between centres of heat collection and release.
Water near 30 °C	Density $\rho \approx 996 \text{ kg/m}^3$; heat capacity $c_p \approx 4180 \text{ J/(kg K)}$; expansion coefficient $\beta \approx 3.0 \times 10^{-4} \text{ /K}$ [26, Table 3]. Assume viscosity $\mu = 0.8 \text{ mPa s}$.
Common pipe route	Assume smooth pipe, $L = 160$ m, fitting-loss factor $\sum K = 20$ [24]. Extra branch and radiator losses are excluded.

The chosen water-temperature differences, 10 °C and 3 °C, test stronger and weaker driving conditions. A smaller difference weakens buoyancy while requiring more flow to carry the same heat load.

Heat rate $\dot{Q}$ is energy carried per second; $\dot{m}$ is water mass flow. Assume steady liquid flow and approximate the hot/cold leg difference by the water temperature change ΔT :

$$\dot{Q} = A_f q'' \approx 1220 \text{ m}^2 \times 100 \text{ W/m}^2 = 122 \text{ kW}, \quad (3.2)$$

$$\dot{m} = \frac{\dot{Q}}{c_p \Delta T}; \quad \Delta p_b \approx \rho g H \beta \Delta T, \quad (3.3)$$

$$\Delta p_{\text{loss}} = \left(f \frac{L}{D} + \sum K \right) \frac{\rho v^2}{2}; \quad v = \frac{4\dot{m}}{\rho \pi D^2}. \quad (3.4)$$

Here D is internal diameter, v water speed and $g = 9.81 \text{ m/s}^2$. Use Darcy friction factor $f = 0.3164/\text{Re}^{0.25}$, with $\text{Re} = \rho v D / \mu$ [24], and solve where friction equals buoyancy [22]. $\text{Re} \approx 26\,000 - 39\,000$ supports the smooth-pipe turbulent-flow approximation.

Table 3.6: Circulation estimates for the 122 kW load.

Water difference	Buoyancy pressure	Water flow	Internal diameter
10 °C (= 10 K)	$\approx 300 \text{ Pa}$	2.9 kg/s	$\approx 180 \text{ mm}$
3 °C (= 3 K)	$\approx 90 \text{ Pa}$	9.7 kg/s	$\approx 400 \text{ mm}$

Reducing the temperature difference more than doubles the required main-pipe diameter. These estimates use the whole available buoyancy pressure, because the diameter was solved for friction equal to buoyancy; radiator, collector and fitting losses need a budget of their own (Section 3.3.5). Varying the assumed heat input from 50 to 150 W/m^2 gives a load of 61–183 kW.

3.2.8 Engineering Check 2 — Roof Heat Release

In steady operation, the roof must release the heat collected by the façade. Raman et al. [27] measured 40.1 W/m^2 net cooling at air temperature under sunlight above 850 W/m^2 . This specialised emitter is used as a benchmark only.

$$A_{\text{release}} = \dot{Q} / q''_{\text{net}} \quad (3.5)$$

For the engineering estimate, assume a 35°C roof, 30°C air, effective sky temperature 20°C , solar reflectance $r = 0.95$, thermal emittance $\varepsilon = 0.90$, sunlight $G = 800 \text{ W/m}^2$ and convection coefficient $h = 3 - -8 \text{ W/(m}^2 \text{ K)}$. Effective sky temperature represents incoming infrared radiation, not air temperature. Neglect back-surface exchange and assume an unobstructed sky:

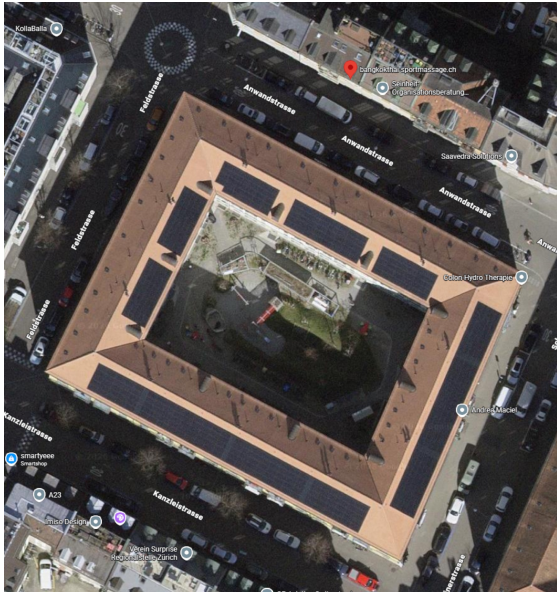
$$q''_{\text{net}} = h(T_{\text{roof}} - T_{\text{air}}) + \varepsilon\sigma(T_{\text{roof}}^4 - T_{\text{sky}}^4) - (1 - r)G. \quad (3.6)$$

The terms represent heat to air, infrared heat exchange and absorbed sunlight. With $\sigma = 5.67 \times 10^{-8} \text{ W/(m}^2 \text{ K}^4)$, net release is $15 - 40 + 83 - 40 \approx 58 - -83 \text{ W/m}^2$. Roof tilt and nearby buildings would change this estimate. This is a daytime case ($G = 800 \text{ W/m}^2$); a night case has not been calculated.

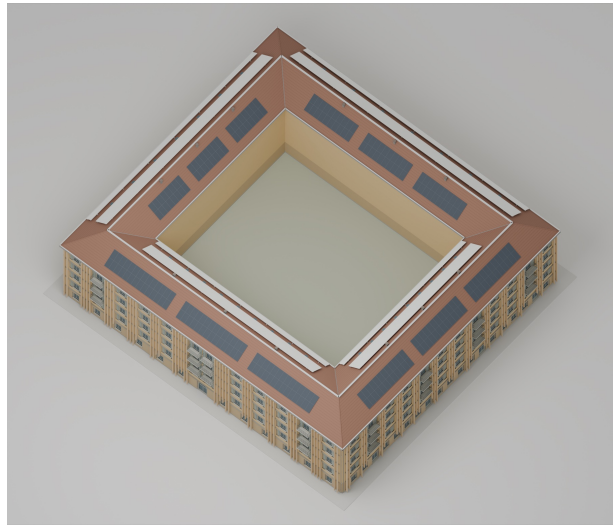
Table 3.7: Roof cooling area required to release the assumed 122 kW load.

Case	Net heat release	Roof area for 122 kW
Published emitter benchmark	40.1 W/m^2	$\approx 3050 \text{ m}^2$
Assumed 35°C roof, 20°C sky	$58 - 83 \text{ W/m}^2$	$\approx 1470 - 2100 \text{ m}^2$
Same roof, warmer 30°C sky	$4 - 29 \text{ W/m}^2$	$\approx 4200 - 29\,500 \text{ m}^2$

The model allocates about 740 m^2 to cooling: 168 m total strip width $\times 4.39 \text{ m}$ combined slope depth. Solar panels, rooflights and gaps limit this area (Figure 3.8). The first two cases give about $30 - 62 \text{ kW}$, supporting roughly $300 - 620 \text{ m}^2$ of façade at 100 W/m^2 . A warmer sky reduces radiative heat loss; near-zero net output makes the required area highly sensitive. The connected façade area must therefore be reduced.



(a) Reference building — Google Maps



(b) Proposed roof layout

Figure 3.8: Reference Block in Modellierungsgebiet 03 [2, pp. 148–151] and our model. Cooling strips share the roof with solar panels and rooflights.

3.3 Concept Evaluation & Iteration

3.3.1 Comparison against the KPIs

Table 3.8 compares the concepts against the KPIs of Table 2.5 using the estimates from Sections 3.2.7, 3.2.8 and 3.2.4; no surface-temperature reduction has been calculated. Each rating is on a 1–5 scale and carries its reason; an unrated KPI is a gap in evidence and does not count as zero.

Table 3.8: KPI comparison using preliminary engineering estimates. Ratings 1–5; “–” means no evidence yet.

KPI (unit)	A V1 — water façade	B V1 — roughness spray
KPI1 solar reflectance (–)	Roof: 0.95 assumed. 4 : high value, but assumed and only on the roof.	Not specified. 2 : no reflective function is designed in.
KPI2 shaded share of the façade (%)	Fins added; duration not calculated. 3 : the function is present, the number is missing.	0% added by the coating. 1 .
KPI3 heat removed (kW; W/m ²)	740 m ² roof: ≈ 30 –62 kW across benchmark / cooler-sky cases; ≈ 3 –22 kW for warmer sky. 4 : estimated, roof-limited.	About 12 kW by the order-of-magnitude estimate of Section 3.2.4; released into the canyon air, and the sign is uncertain at low wind. 2 .
Heat transport (kW; kg/s)	122 kW assumed load requires 2.9 kg/s at 10 °C difference or 9.7 kg/s at 3 °C.	Local exchange; rate not quantified.
KPI4 surface temperature reduction (K)	Not calculated. –	Not calculated. –
KPI5 water consumed (L/(m ² d))	0 in normal operation; filling and leakage are separate. 5 .	0 for a dry coating. 5 .
KPI6 performance retained (%)	No ageing data. –	No ageing data. –
Sum of rated KPIs (max. 20)	16	10

The flow rates are requirements that the circulation check has to meet. Even the highest roof estimate is below the assumed 122 kW load.

3.3.2 Engineering Review and Selection

We reviewed both V1 concepts in the team and with an AI assistant as critical reviewer; the tool is declared in Appendix A. Table 3.9 lists the issues raised, our response and the resulting decision.

Reduced air exchange in Modellierungsgebiet 03 makes B’s airflow-dependent benefit uncertain [2, pp. 148–151], [23]. Elephant hairs can aid heat transfer at low wind, whereas transverse wooden ribs reduced natural convection in another study [7], [25].

A was selected because it combines shading with heat transport to a separate release surface, and because B’s benefit is not demonstrated (Table 3.8).

3.3.3 Final Concept V2 — Construction and Heat Paths

V2 moves the pipes into separable panels outside the original wall, allowing both new construction and retrofit. The wall is retained apart from mounting connections. A small gap behind the

Table 3.9: Team review and critical assessment of AI feedback.

Review issue	Response	Resulting decision
Pipes hidden behind the absorber risk hidden damage	Accept	Move pipes into a separate external layer; enable access and retrofit.
Roof area and resistance limit A	Accept	Limit connected area; use larger mains and short parallel circuits.
Weak ventilation rules out B	Qualify	Airflow is uncertain, but texture does not automatically fail. Test its mechanism.
Existing KPI definitions are incomplete	Accept	Use heat-flow units; define targets and test conditions before measurement.

added layer is intended to let escaped water fall away externally, reducing hidden wall-damage risk without guaranteeing leak-proof protection (Figure 3.9).

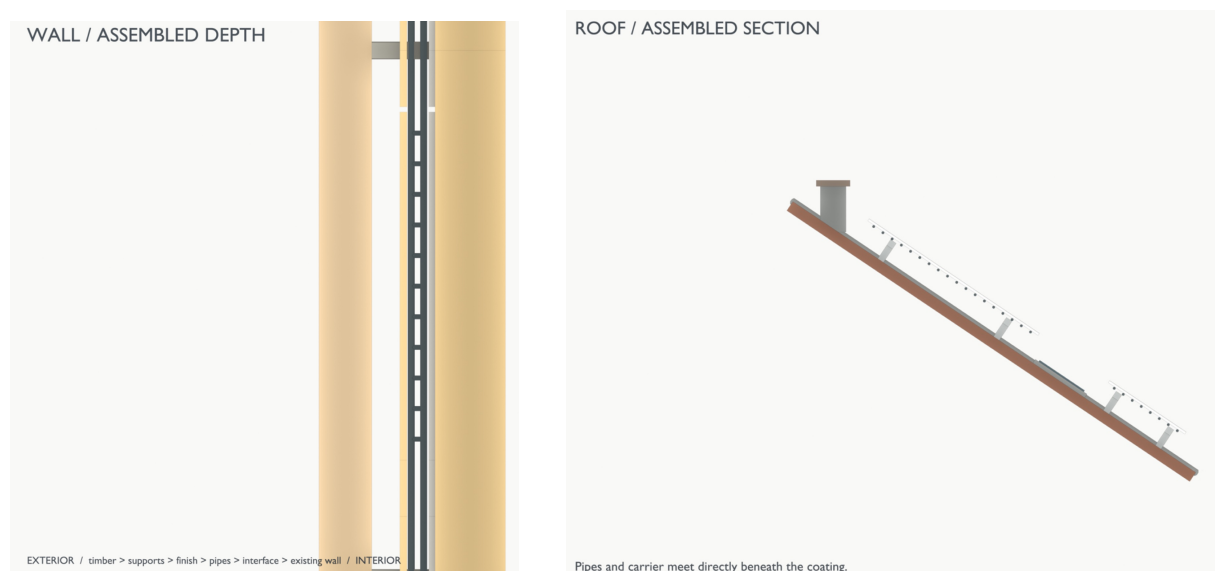


Figure 3.9: Our assembled sections. Façade, outside inward: timber ribs, supports/air space, collector panel, pipes, interface, existing wall. Roof: coating, carrier plate, pipes, supports, existing roof. Render: project team.

Heat passes from the façade panel into water, then from the roof pipes into the carrier plate and coated surface. The Saharan silver ant inspires solar reflection and infrared emission [10]; the assumed 95 % reflectance still needs material verification. No increase in convection is assumed; convective heat release requires a surface warmer than the air (Figure 3.10).

3.3.4 Pedestrian Space and Maintenance Access

Compared with V1, we cut back the timber fins to start at the first floor, about 3.4 m above ground (Figure 3.11). A new building could set its wall back to accommodate the fins; this retrofit must retain the wall beside the pavement. Reducing the modelled ground-floor projection from 0.66 m to 0.20 m leaves 0.46 m more space.

Removing individual panels exposes the pipes for inspection and repair without opening the original wall (Figure 3.12). This improves access compared with V1's hidden pipes. Rib spacing, winter gains and the material checks are taken up in Section 4.2.

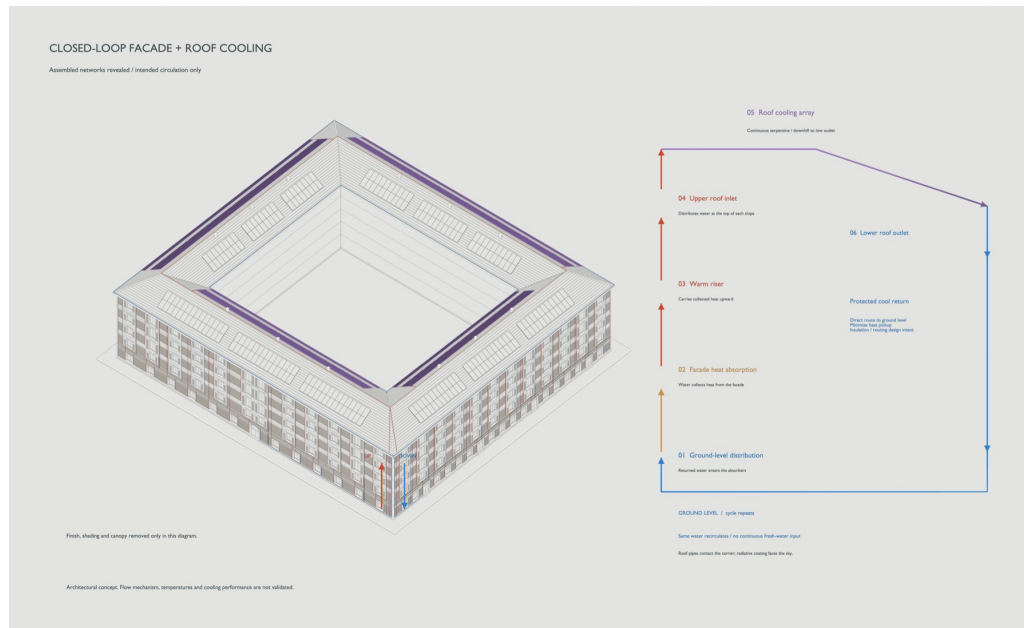


Figure 3.10: Intended heat-transfer loop: façade collection → warm riser → roof heat release → cool return. Render: project team.



(a) Fins start at the first floor



(b) Thinner base beside the pavement

Figure 3.11: Revised timber fins and pedestrian space in our model. Dimensions are design estimates. Render: project team.

3.3.5 Updated Engineering Assessment

For an illustrative roof output of 60 W/m^2 , the 740 m^2 cooling area releases about 44 kW , supporting roughly 440 m^2 of façade at 100 W/m^2 . Repeating the circulation check with the same height and common route gives Table 3.10.

Table 3.10: Pipe-size estimates after reducing the connected heat load.

Water difference	Buoyancy pressure	Water flow at 44 kW	Internal diameter
10°C (= 10 K)	$\approx 300 \text{ Pa}$	$\approx 1.1 \text{ kg/s}$	$\approx 120 \text{ mm}$
3°C (= 3 K)	$\approx 90 \text{ Pa}$	$\approx 3.5 \text{ kg/s}$	$\approx 255 \text{ mm}$

The drawn roof uses four long coils. Area divided by 0.16 m pipe spacing gives about $740/0.16 \approx 4620 \text{ m}$ of pipe, or 1150 m per coil. Assuming 40 mm internal diameter, equal flow and

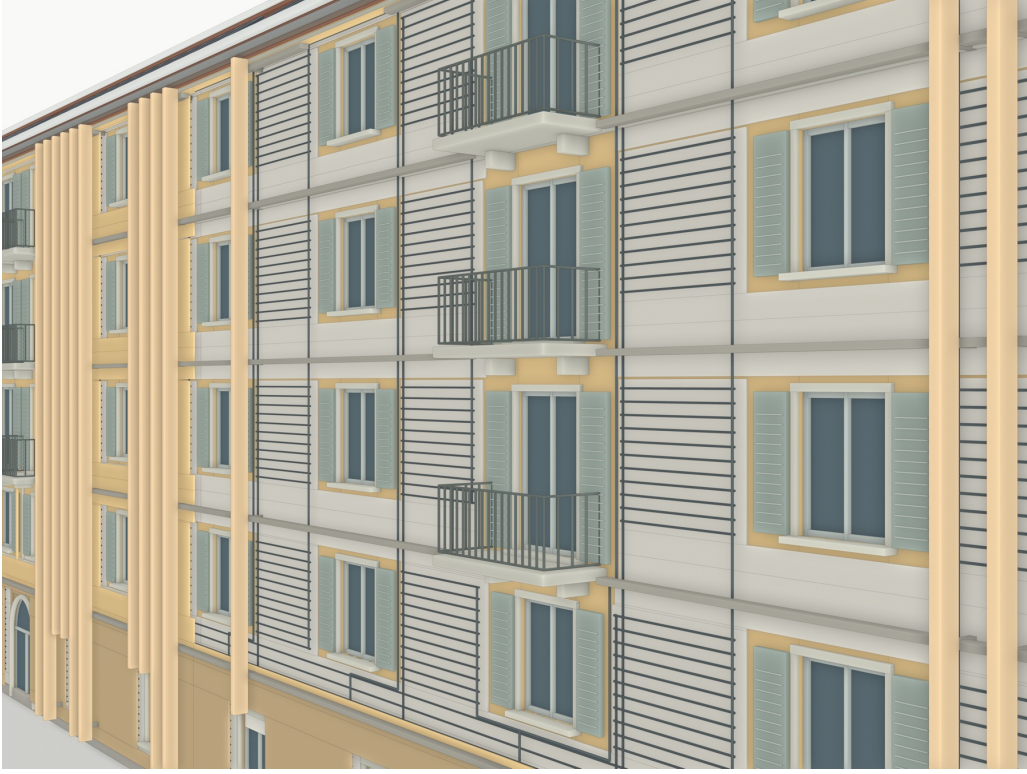


Figure 3.12: Pipe-access detail from our model (view 17). Parts of the covering are removed to reveal the pipes behind the proposed separable façade panels. Render: project team.

a 10°C water difference, each coil carries $\dot{m} = 1.05/4 \approx 0.26 \text{ kg/s}$, so $v \approx 0.21 \text{ m/s}$, $\text{Re} \approx 10\,500$ and $f \approx 0.031$; the Darcy equation (3.4) gives

$$\Delta p_{\text{coil}} \approx 0.031 \cdot \frac{1150}{0.040} \cdot \frac{996 \cdot 0.21^2}{2} \approx 20 \text{ kPa}$$

loss per coil at 44 kW total load, excluding bends. This exceeds the 0.30 kPa buoyancy pressure by roughly a factor of 67: the long-coil arrangement must change.

The roof and water-flow estimates prescribe temperatures separately. Water temperature, flow and pipe-to-surface heat transfer must be solved together before assigning an operating output.

A pressure budget. A thermosiphon works only if the sum of all losses around the loop stays below the buoyancy pressure. For the 10 K case we split the 300 Pa as follows: mains at most 30 Pa, roof circuits at most 60 Pa, façade panels, headers and fittings at most 150 Pa, and about 60 Pa in reserve for start-up. The mains of Table 3.10 used the whole budget, so they grow: with $\dot{m} = 1.05 \text{ kg/s}$, $L = 160 \text{ m}$ and $\Sigma K = 20$, an internal diameter of 200 mm gives $v \approx 0.034 \text{ m/s}$, $\text{Re} \approx 8400$, $f \approx 0.033$ and a loss of about 26 Pa, inside the mains share.

Parallel roof circuits. We keep the total roof pipe length and split it into N parallel circuits of length $L_c = 4620 \text{ m}/N$, each carrying $\dot{m}/N$, fed from a supply header and collected by a return header. The loss per circuit, which is the loss of the whole roof stage because the circuits are in parallel, is

$$\Delta p_{\text{roof}} = f \frac{L_c}{D} \frac{\rho v^2}{2}, \quad v = \frac{4\dot{m}}{N\pi D^2}, \quad f = \frac{64}{\text{Re}} \text{ for } \text{Re} < 2300, \quad (3.7)$$

with the Blasius correlation above 2300 [24]. In the laminar range Δp_{roof} scales with vL_c , so doubling N divides the loss by four. Table 3.11 evaluates Equation (3.7) for $D = 40 \text{ mm}$, $\rho = 996 \text{ kg/m}^3$, $\mu = 0.8 \text{ mPa s}$ and the 10 K flow of 1.05 kg/s .

Table 3.11: Roof stage as N parallel circuits, 40 mm internal diameter, total flow 1.05 kg/s (44 kW at 10 K). Budget for the roof stage: 60 Pa.

N	L_c (m)	$\dot{m}$ per circuit (kg/s)	v (m/s)	Re	f	Δp_{roof} (Pa)
4	1150	0.26	0.21	10 500	0.031	$\approx 19\,700$
16	289	0.066	0.052	2600	0.044	≈ 440
32	144	0.033	0.026	1300	0.049	≈ 60
64	72	0.016	0.013	650	0.098	≈ 15

With 32 to 64 circuits of 70–145 m, the roof stage stays inside its 60 Pa share, and the whole loop closes on paper: about 26 Pa in the mains, 15–60 Pa on the roof, up to 150 Pa for panels, headers and fittings, and a reserve. Two consequences follow. The flow in the roof circuits is laminar, which lowers the heat transfer from the water to the pipe wall and the carrier plate, so a low pressure loss is bought with a lower plate temperature; the measured roof output asked for in Table 3.12 is what settles this. And the 3 K case does not close with these sizes: its 3.5 kg/s would cost about 250 Pa in the 200 mm mains alone against 90 Pa available. A thermosiphon answers such a shortfall by running at a lower flow and a larger temperature difference, and whether it starts up at all from a small difference is the first test in Table 3.12. The long coils in Figure 3.10 are superseded by this network; the drawing has not yet been updated.

Roof-pipe water volume is approximately $4620 \text{ m} \times \pi \times (0.04 \text{ m})^2 / 4 \approx 5.8 \text{ m}^3$, excluding façade pipes and headers, which adds about 5.8 t to the roof load.

Winter operation. Before frost, the proposed circuit is emptied into storage and refilled when needed. Drainback requires sloping pipes and air entry; Baechler and Love [28, Ch. 3] give about 2.1 % slope. Trapped water and pump-assisted refill remain checks [28].

3.3.6 Priorities for Further Development

Table 3.12: Ranked checks needed to develop Concept V2.

Priority	Check and design decision
1 Circulation	Size the parallel network including all losses (Table 3.11); test start-up at small temperature differences. Decide whether buoyancy is sufficient.
2 Roof output	Measure heat release under varied sun, wind and sky conditions. Set the connected façade area from the result.
3 Winter operation	Test emptying and refill of one representative branch; measure retained water before freeze–thaw testing.
4 Panel assembly	Check pipe contact, leak behavior, removable connections and mounting loads.
5 Building and materials	Check pedestrian clearance, heritage compatibility, glare, winter gains, weathering, chemical safety and repair.

4 Sustainability & Life's Principles

4.1 Life's Principles applied to Concept V2

4.1.1 Be resource efficient

Cooling needs no auxiliary electricity: circulation is driven by buoyancy alone (Section 3.2.7), and the loop is closed, so no water is consumed in operation (Table 3.8).

Pumpless operation is not demonstrated. The mains alone use the whole ≈ 300 Pa buoyancy pressure, the drawn roof coils exceed it by a factor of about 67, and refilling after drainback may need a pump (Section 3.3.5).

4.1.2 Adapt to changing conditions

The daily cycle drives the system: a hotter façade produces a larger density difference and more flow, so cooling rises and falls with the load without a controller (Section 3.2.7), and the roof is expected to release heat to the night sky, although only the daytime case of Section 3.2.8 has been calculated. The seasonal cycle is met by draining the loop before frost and refilling it in spring (Section 3.3.5).

The seasonal adaptation is a withdrawal rather than an adaptation. Even after draining the pipes there is a remaining risk of freezing leftover water. Furthermore the ribs and the reflective roof act the same in winter.

4.1.3 Use life-friendly chemistry

The heat-transfer fluid is plain water. Frost protection is solved by draining the pipes instead of anti freeze such as glycol. Reflection and emission are sought through the surface optics of the silver ant [10]; the white roof coating owes its reflectance to a titanium dioxide pigment, which is a functional additive (Section 4.3).

We meet this principle by leaving substances out; we have not verified the ones we use. Two points remain open and unchecked: the timber treatment of the ribs and the binder and preservatives of the roof coating.

4.2 Environmental benefits, limitations and trade-offs

4.2.1 Environmental benefits

The climate benefit is what V2 replaces: every hour of air conditioning it makes unnecessary is avoided emissions and avoided waste heat in the street (Section 2.1). Radiation to the night sky carries the collected heat out of the urban air; the other release options of Table 3.3, exchange with the air beside the wall or evaporation, leave it in the street (Section 3.2.8). The street itself should be cooler: the shaded façade radiates less onto the pavement, there is no condenser at pedestrian height, and reflection happens on the roof, where it does not raise heat stress in the canyon the way a reflective wall does [21]. The retrofit is reversible: nothing is demolished; the panels can be removed and the building returned to its previous state, so the retrofit creates no demolition waste and consumes no land. In operation the system consumes no water and contains no refrigerant; what it does release is the abrasion of the roof coating into the run-off (Section 4.3).

4.2.2 Limitations

The export of heat depends on a clear, dry night sky. In the humid, hazy nights of a heat wave most of the heat only warms the roof air, so the concept helps least in the conditions it is meant for (Section 3.2.8). Outside the summer the drained loop does nothing, and no other

environmental function fills the gap. No cooling figure exists, so the avoided emissions cannot be counted, and the whole benefit rests on circulation without a pump, the first open check of Table 3.12. The added materials carry embodied emissions that the constraint in Table 2.1 asks to be repaid within the service life; we carried out no life cycle assessment, so neither side of that comparison exists.

4.2.3 Trade-offs

Ribs and panels change how the building looks; on a listed façade this collides with heritage protection [2, pp. 148–151]. What shades and reflects in summer also rejects winter sun and can raise heating demand and its emissions [29]; the net annual effect is unknown. Every square metre of roof that releases heat is a square metre without photovoltaic yield (Figure 3.8). Avoiding antifreeze costs the shoulder seasons and admits air that may bring back a corrosion inhibitor (Section 3.3.5). Repairable external pipes are the same visible layer that alters the façade.

4.3 Materials, durability, repair and circularity

The timber ribs are worn by UV, driving rain and freeze–thaw, so they belong to the biological cycle. What weathers off lands on the pavement, and the timber must be untreated, naturally durable, or protected by detailing rather than by chemistry. The collector panels and their finish see slow surface wear and are otherwise used as structure: technical cycle, if the panel is one recoverable material with a benign finish. The panels are designed to be removable (Figure 3.12). The roof coating is worn by UV, rain, soiling and cleaning, and abraded into run-off that reaches the stormwater network of Section 2.3, so it must break down harmlessly. Pipework, carrier plate, supports and fixings are used, not consumed: technical cycle, recoverable if specified as separable single materials. The water in the loop is used but mobile and fits neither cycle cleanly: about 5.8 m³ is drained to storage each winter and eventually disposed of, and with an inhibitor it is the one component that can leave the system by accident.

Measured against the eight cycle design principles of the lecture [30], V2 answers four in principle and none with evidence. Modular construction (3) and reparability (5) follow from moving the pipes out of the wall: a panel can be removed and the façade pipes inspected without opening the wall (Sections 3.3.2 to 3.3.4). Ecotoxic chemicals (1) and recyclable materials (2) are answered above.

Material checked so far. White cool-roof coatings owe their reflectance to titanium dioxide, so it was checked in the REACH database of the European Chemicals Agency [31]. The powder form was classified in 2020 as a suspected carcinogen by inhalation (Regulation (EU) 2020/217, H351) [32]. The General Court annulled the classification in 2022 and the Court of Justice confirmed the annulment on 1 August 2025 [33], so the pigment currently carries no harmonized hazard classification. The concern was dust inhalation, not the applied film, and the ruling turned on the reliability of one study rather than on a finding of safety. The binder and the film preservatives are what reach the run-off instead. One substance of at least four has been checked. The other three belong to item 5 of Table 3.12.

4.4 Did sustainability considerations change the design?

Yes, in four instances documented in Sections 3.2 and 3.3.

1. **The reflective surface moved from the façade to the roof.** Concept C reflected on the façade and was set aside because reflective walls raised pedestrian heat stress in simulations of a Stuttgart street canyon [21]; Concept A reflects on the roof instead (Section 3.2.2). The change was made for the people at street level.
2. **Water and power demand removed two options before any concept was drawn.** The evaporating wet layer and the fan-driven cavity of Table 3.3 were not developed further because of their water and power demand (Section 3.2.2). The Energy and Water constraints of Table 2.1 decided this.
3. **The pipes moved out of the wall.** V1 hid them behind the absorber layer without access. The engineering review found that in case of damage the lack of access was unacceptable (Table 3.9), and therefore V2 placed them in separable external panels with a drainage gap behind them (Section 3.3.3). The recorded motive was repairability; the material separation is a consequence.
4. **Water instead of glycol.** Frost protection was decided by drainback (Section 3.3.5); keeping an additive out of the loop is a consequence of that choice.

The winter heating balance and the embodied emissions did not change the design; they are constraints in Table 2.1 that remain unquantified.

4.5 Limits of this assessment

The assessment is made against Concept V2 as described in Section 3.3.3, whose central claim of buoyancy-only circulation is the first-ranked open check of Table 3.12; if a pump is needed, the Energy constraint and the resource-efficiency assessment of Section 4.1 fall with it. Cascade use and return logistics lie outside the project boundary, and of the substances the concept would put into the street one has been checked and three have not.

5 Discussion & Conclusion

5.1 Contribution of the biomimicry process

Biomimicry helped us explore cooling functions before committing to familiar building solutions. Studying how organisms manage heat encouraged us to abstract and combine strategies, leading to the proposed link between façade shading, fluid transport and a roof emitter (Sections 3.1–3.2). Without this process, we would probably have focused on improving existing solutions individually.

5.2 Strengths and weaknesses of the final concept

V2 offers an unconventional approach for new buildings and especially retrofits: shading is intended to limit façade heating, while the water loop transfers collected heat to the roof. This aims to reduce heat release at street level (Section 3.2.2). However, circulation and the available roof area are the main limitations. The first roof layout exceeded the available driving pressure and was replaced by a parallel network (Section 3.3.5). Pump-free circulation remains unverified, and refilling may require a pump.

5.3 Assessment of the challenge and the KPIs

The KPIs guided design decisions: the heat-flow estimates limited the connected façade area (Section 3.2.8). Against Table 3.8, KPI5 is met on paper, KPI1 and KPI3 are estimates, and KPI2, KPI4 and KPI6 have no evidence yet. Reflectance, shading duration, durability and actual cooling performance still need verification.

5.4 Innovative character and limits of the evidence

The innovative aspect is the combination of biological principles in a maintainable façade system intended for new buildings and retrofits. Our main lesson from the course is that biological ideas become more useful when calculations and practical constraints shape their technical transfer. Circulation and roof-output tests (Table 3.12) are the priorities for assessing the concept's potential.

6 References

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Appendix A – AI Use Statement

Table A.1 lists the AI-supported activities in this project, what each contributed and how we checked the output. No factual claim in the report is cited to an AI tool; every biological and technical statement carries a reference of its own.

Table A.1: AI use statement.

AI tool	Used for	How the output influenced the project	How important output was verified
AskNature website and AskNature Chat	Searching biological strategies for functions F1–F5 with the keywords and biologized questions of Section 3.1.1	Produced the candidate list of Table 3.1, including the models we later de-selected	Each strategy page was opened and read. Where a primary paper exists it is cited in Section 3.1.2 instead of the AskNature page (silver ant, cactus, camel, snail, pine cone, honeybee); the AskNature pages remain cited as the lead for the elephant models and the Morpho
Large language model assistant (Claude, Anthropic)	Structuring the scoping notes; drafting and editing report text; converting the report to L ^A T _E X; an editorial audit of the draft against the course rubric (repetition, unsupported claims, citation gaps); recomputing the engineering estimates	Draft wording in Sections 1–5 and the layout of most tables; the audit produced the list of corrections applied in this version (figure numbering, missing references, KPI definitions, removal of repeated passages)	The team decided on every audit finding. All numbers in Sections 3.2.7, 3.2.8 and 3.3.5 were recomputed by hand from the stated inputs before the text was accepted. Any claim the assistant proposed without a source was either given a source we read ourselves or removed
AI as critical reviewer	Reviewing Concept A V1 and B V1 with the Engineering Canvas, the functions and the KPIs (Section 3.3.2)	Four review issues in Table 3.9; three were accepted and changed the design (pipes into external panels, reduced connected area, KPI units), one was qualified	Each issue was assessed by the team against the literature named in Section 3.3.2; the qualified issue (ventilation) was not accepted because the cited studies disagree
Image generation	[TODO: State whether any figure was AI-generated (for example the concept renders of Figures 3.6 and 3.10); if so, name the tool]	[TODO: Illustration only; no figure is used as measurement or evidence]	[TODO: Each generated image was compared with the hand sketch it was made from]

Appendix B – Optional Supporting Material

B.1 Glossary

Albedo Refers to an index from 0 – 1 on how much light is reflected.

PV Photovoltaic

Radiation Shortwave radiation is the type of radiation emitted by the sun. Longwave radiation is thermal radiation which is emitted from a surface into the surrounding atmosphere.

UHI Urban Heat Island

UV-Light Ultraviolet Light

B.2 Decision log

Table B.1 lists the decisions that shaped the concept, the alternative each displaced, and where in the report the reason is given.

Table B.1: Decision log.

Decision	Chosen	Rejected alternative	Reason and where it is argued
Design question	Reduce the heat island by changing façade and roof of existing buildings	Three broader or wrongly framed candidates	Anchored to the system boundary and testable; Section 2.4
Reference city and site	Zurich, Modellierungsgebiet 03	A generic city or a freely chosen building	Documented heat map and albedo classes, open geodata; Section 2.2
System of interest	Façade and roof of the building envelope	The whole building, internal cooling	Keeps the project finishable; Table 2.2
KPI units	Physical units with baseline and target	Temperature labels without targets	Needed for a defensible comparison; Section 2.7
Reflective surface	On the roof	On the façade (Concept C)	Reflective walls raised pedestrian heat stress in canyon simulations; Section 3.2.2
Water and power	No evaporating layer, no fans	Wet layer, fan-driven cavity	Water and Energy constraints of Table 2.1; Section 3.2.2
Concept pair	A (water façade) and B (textured coating)	C (responsive louvers)	Humidity is not a cooling signal; Section 3.2.2
Final concept	A, revised to V2	B	B's benefit is unproven and of uncertain sign; Section 3.3.2
Pipe location	Separable external collector panels with a drainage gap	Absorber layer without access (V1)	Hidden damage and no repair access; Section 3.3.3
Fins at ground level	Start at the first floor, thinner base	Fins to the ground (V1)	Pavement width and door operation; Section 3.3.4
Roof network	Many short parallel circuits	Four long coils	Long coils lose about 67 times the available buoyancy pressure; Section 3.3.5
Frost protection	Drainback	Glycol	Keeps an additive out of a loop on an occupied façade, at the cost of shoulder-season operation; Sections 3.3.5 and 4.2